

Livestock Sector Forecasting in Kenya: A Cohort-Based Quarterly Analysis on Population, Production, and Feed Demand

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Abstract—

Kenya's livestock sector contributes 42% of agricultural GDP and 12% of national GDP, yet quantitative planning remains constrained by the absence of reproducible, data-driven analytical models. This study presents a cohort-based quarterly forecasting framework spanning nine value chains: dairy cattle, beef cattle, camels, goats, sheep, poultry, pigs, rabbits, and apiculture, initialized from the 2019 KNBS livestock census with projections from 2020 to 2041. The model applies a quarterly transition sequence covering mortality, culling, offtake, age-based maturation, and reproduction across all cohorts. Production modules translate herd and flock dynamics into species-specific output metrics, while a carrying capacity constraint — derived from satellite-based vegetation indices and adjusted by a utilization factor — regulates growth against forage availability. A feed demand module estimates dry matter requirements by species and agro-ecological zone. Projected outputs indicate a CAGR of 3.6-6% across all value chains. Beyond forecasting, the framework exposes productivity bottlenecks and structural inefficiencies, including feed deficits, low offtake rates, and suboptimal reproductive performance, that aggregate assessments typically obscure. The model provides an analytical basis for food security planning, targeted infrastructure investment, and evidence-based livestock policy.

Index Terms—Livestock model forecasting, carrying capacity, biological parameters

I. INTRODUCTION

In most African countries, the livestock sector provides a living for millions of rural communities and contributes significantly to poverty reduction efforts. In Kenya, the sector provides jobs across value chains, generate revenue and contributes to food security for both rural and urban populations, making a significant economic contribution [?]. The livestock industry significantly impacts Kenya's food system, accounting for over 12% of the nation's Gross Domestic Product (GDP), 40% of agricultural GDP, and 22% of food system GDP [?]. By 2050, there will be about 96 million people living in

Kenya, with almost half of them living in cities [?]. Due to population growth, urbanization, rising affluence, and shifting dietary preferences, demand for animal-source foods is expected to rise significantly [?]. Sustained increases in livestock production, effective management of feed and water resources, and evidence-based planning to support the livestock sector's sustainable development are all necessary to meet this rising demand. Therefore, evidence-based planning, strategic investment, and the development of a sustainable livestock sector depend on accurate forecasting frameworks that predict livestock populations, production, feed demand, and resource requirements. Statistical time-series models are often used to forecast animal numbers based on past data. These approaches anticipate future livestock numbers based on past population trends and have been used to aid in livestock planning and management. In [?], autoregressive and probabilistic time-series models were used to forecast pig populations, while [?] compared autoregressive integrated moving average (ARIMA) and artificial neural network (ANN) models for forecasting cattle populations in Türkiye. Similarly, [?] used exponential smoothing to forecast beef cattle population in Indonesia. While these approaches can effectively capture historical patterns and produce useful forecasts, they are mostly based on past trends and do not explicitly account for biological processes such as reproduction, mortality, maturation, culling, or offtake. As a result, their ability to assess the impact of biological characteristics and management interventions on future livestock population dynamics is restricted.

In contrast to statistical forecasting methods, biological simulation models represent the demographic dynamics that drive livestock populations. Animals are divided into cohorts based on their age, sex, or production stage, with transitions caused by biological processes such as reproduction, maturation, mortality, culling, and replacement. These models

provide a biologically realistic picture of herd dynamics, allowing the evaluation of alternative management options. To simulate herd dynamics, [?] created an age- and sex-structured cattle population model that takes into account management techniques and animal migrations. Such models are useful decision-support tools because they link biological processes to herd structure and productivity. Existing biological models therefore provide valuable representations of herd dynamics, but a gap remains for replicable frameworks that jointly forecast population dynamics, production, feed requirements, and resource constraints across multiple livestock value chains. To address these shortcomings, this study developed a cohort-based quarterly forecasting framework for Kenya's nine major livestock value chains. The framework explicitly modeled biological and management processes, including reproduction, mortality, maturation, culling, and offtake, and links livestock population dynamics to species-specific production, feed demand, and environmental carrying capacity across agro-ecological regions. By integrating these components, the framework enabled consistent projections of livestock populations, production, and resource requirements, providing a decision-support tool for evidence-based livestock planning and policy development in Kenya.

II. RELATED WORK

Write related work (literature review) here.

III. METHODOLOGY

Write methodology here.

IV. RESULTS

Write results here.

V. DISCUSSION

Write discussion here.

VI. CONCLUSION

Write conclusion here.

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Write acknowledgment here.

REFERENCES